

Smart Grid technology by Ecolibrium

You can now monitor your electrical appliances power consumption in real time with a mobile or an internet connection

Microsoft loves open source

Microsoft's General Manager quoted that criticism on open source by Microsoft was a mistake

Tomorrow's tech

hardware that senses the direction and force of bullet fire, explosions etc. The force feedback is perfectly synchronised with the screen. Alternate Reality Games use wearable computers for more fun and thrill.

Managing personal data

If you wear them, they can be no less than your secretaries. They can store your personal data and assist you in managing stuff which is crucial for you. Accenture has come up with a "Personal Awareness Assistant" that passively listens to what the user says by using a recording device that has an ability to respond to particular contexts and situations. The device uses a speech recognition engine, an inconspicuous camera, a scrolling audio buffer and 2 microphones. For example, when you introduce yourself to a new person, this device records the name of the person, takes a low resolution picture and stores the data and categorises the information in a pre-designated domain. All this is done without you giving any command. This acts like an address book with audio, digital image, date/time and location also. "Personal Awareness Assistant" can also help you shop. You can create a shopping list by speaking a reminder into the device and then access it when you shop.

Exploring and documentation

Wearable computers have been excellently used in exploration, navigation and documentation of information gathered during exploration. Japanese engineers have created a prototype device that clubs the GPS navigation technology into everyday glasses called 'Wearable Personal Navigation System'. The device contains a magnetic directional sensor, a microcomputer and LEDs. The LEDs change their color and placement to guide you the direction. For example, you are all set to go towards North, but you are facing South, then an LED will be lit up indicating that you need to go in the opposite direction.

An Indian geographical scientist Prayas Abhinav along with his team has made shoes called "Bhatka Bhatka". These shoes help you get lost and explore your city. They have a GPS, some Java code, an LED and a vibrator. The shoe recognises and saves the places you have been to already in a particular city. You visit the same place again, it gives a signal as a reminder that you have been here before. Prayas states that this project would help people in exploring different parts of a city.

Security

If these wearable computers can do so much for you, why can't they help you during emergencies! Tanizawa, a Japanese safety equipment manufacturer has developed a U-Met Rescue Helmet that



Hi-tech security systems can monitor in all directions

could be your life saver when needed.

The device has a QVGA camera, a GPS receiver and a cell radio to send images and locational data from emergency workers to a central station every 30 seconds. It vibrates indicating potential dangers. This helmet help you in emergencies to document the scene and seek help from authorities. The helmet is recommended to be used by security personnel and fire fighters.

Military

Military uses wearable computers in the form of modular infantry combat suite,

wearable intelligent weapons and path guiding equipment. French militants use FELIN, a combat suite that offers efficient ergonomics and weight saving. The garment's structure is modular, allowing the soldier to adjust the weight of the suite according to the mission. The suite is also provided a special nuclear, radiological, bacteriological and chemical protection configuration. The suite has a day- night camera, an osteo-phone and a high-speed wearable computer with open universal port interfaces that control energy and links with various components. The system is connected to a man-machine interface enabling the soldier to control the system.

Health

Wearable computers have become a part of medicine and cure in the form of health monitors. A device called "Wrist Pulse Oximeter" is used to record the data like pulse rate and transfer it to your computer for you to analyse it. We now have wearable devices that maintain and monitor body clock to prevent jet lag and disorders in body clocks. We have wearable multi-sensor heart rate monitors that keep a check on the health of heart patients.

So as you have seen, the possibilities with wearable computers



Monitor your heart rate and blood pressure

are endless. And if John Underkoffler, Minority Report science adviser and inventor, is to be believed, the futuristic interface used in the movie is not just possible, but already exists. **d**